BLUE TEAM CAPSTONE · WEEK 1

Mini-SOC
in a box

Building the Lab and Deploying Wazuh

The foundation of security monitoring for LabShop

3	4.14.7	2	5/5
virtual machines	Wazuh version	active agents	test events

Field	Details
Project	LabShop Mini-SOC — 12-week Blue Team Capstone
Week	1 — Build the lab and deploy Wazuh
Platform	VMware Workstation · Ubuntu Server · Windows 10 · Docker
Prepared by	_____
Date	October 2026

Every requirement is evidenced by its code from the brief (L, W, A, V, S, D).

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Requirement codes

Code	Meaning
L	Lab environment — VMs, network, clocks, Docker, snapshots
W	Wazuh platform — stack, dashboard, password, persistence, memory
A	Agents and telemetry — Linux, Windows, Sysmon, web server
V	Verification — deliberately caused events, event path
S	Security and hygiene — no secrets in Git, isolated lab
D	Documentation — README, repository, journal

1 Project overview

LabShop is a small online shop: one web server, one staff PC and no security monitoring at all. This week, as LabShop's security team, I built the foundation: an isolated lab, Wazuh 4.14.7 running in Docker, and two active agents.

1.1 Lab map

Machine	Role	IP	RAM	vCPU	Disk	Network
siem-host	Wazuh (Docker)	10.10.10.10	16 GB	6	256 GB	VMnet2 + NAT (VMnet8)
ubuntu-web	Linux endpoint, Apache, SSH	10.10.10.50	2 GB	2	40 GB	VMnet2
win-client	Windows endpoint, Sysmon	10.10.10.80	4 GB	2	80 GB	VMnet2

The lab network is VMnet2 (host-only, 10.10.10.0/24). The VMs use 22 GB of RAM in total, leaving 10 GB for the Dell. The endpoints had NAT only while software was being downloaded; it was removed afterwards.

1.2 The four Wazuh components

Component	What it does	Where it runs
Agent	Collects logs on the endpoint and sends them to the manager	ubuntu-web, win-client
Manager	Decodes logs, applies rules, generates alerts	siem-host (Docker)
Indexer	Stores alerts and makes them searchable	siem-host (Docker)
Dashboard	Web console that displays the alerts	siem-host (Docker), port 443

Ports to know: **1514** (events), **1515** (agent enrolment), **55000** (API), **443** (dashboard).

1.3 SIEM, EDR and XDR

Term	Plain explanation	Where Wazuh fits
SIEM	A system that collects logs from many sources, correlates them and raises alerts	Manager + Indexer + Dashboard perform this role
EDR	A tool that watches processes and files on the endpoint and can respond	The agent provides endpoint visibility, deeper with Sysmon
XDR	An approach that unifies endpoint, network and cloud data in one place	Wazuh presents itself as a SIEM and XDR platform

2 Lab environment L1–L5

Three VMs were built with the required resources. They reach each other over static IPs, their clocks are in sync, Docker is ready, and every VM has snapshots.

L1 VMs and resources

The three VMs were built in VMware Workstation. siem-host runs with 16 GB RAM, 6 vCPUs and a 256 GB disk. It has two adapters: VMnet2 (lab) and VMnet8 (NAT).

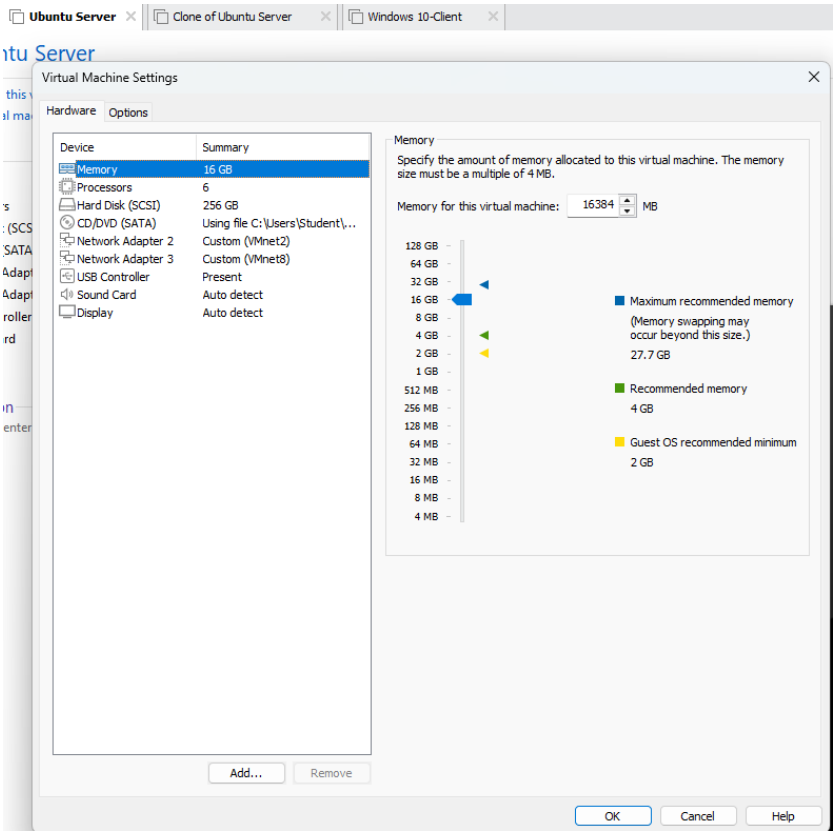


Figure 1. *siem-host VM settings: 16 GB RAM, 6 processors, 256 GB disk, VMnet2 + VMnet8*

ubuntu-web has 2 GB RAM, 2 vCPUs and a 40 GB disk. The screenshot was taken during setup, when the NAT adapter was still attached.

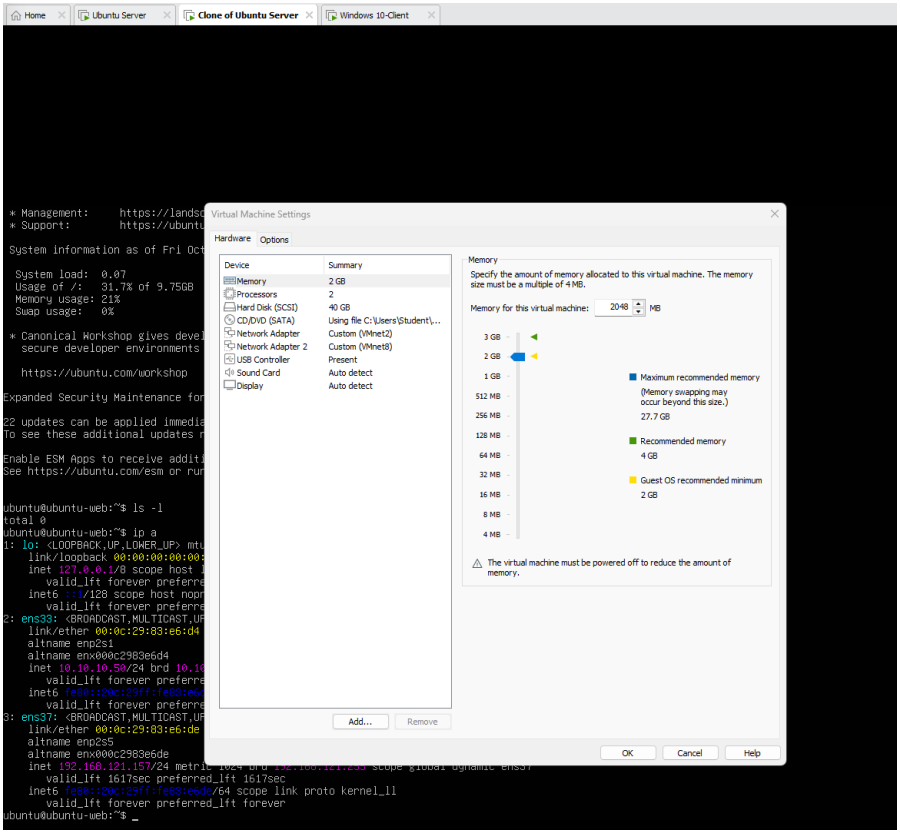


Figure 2. *ubuntu-web VM settings: 2 GB RAM, 2 processors, 40 GB disk (setup stage)*

win-client runs Windows 10 with 4 GB RAM, 2 vCPUs and an 80 GB NVMe disk.

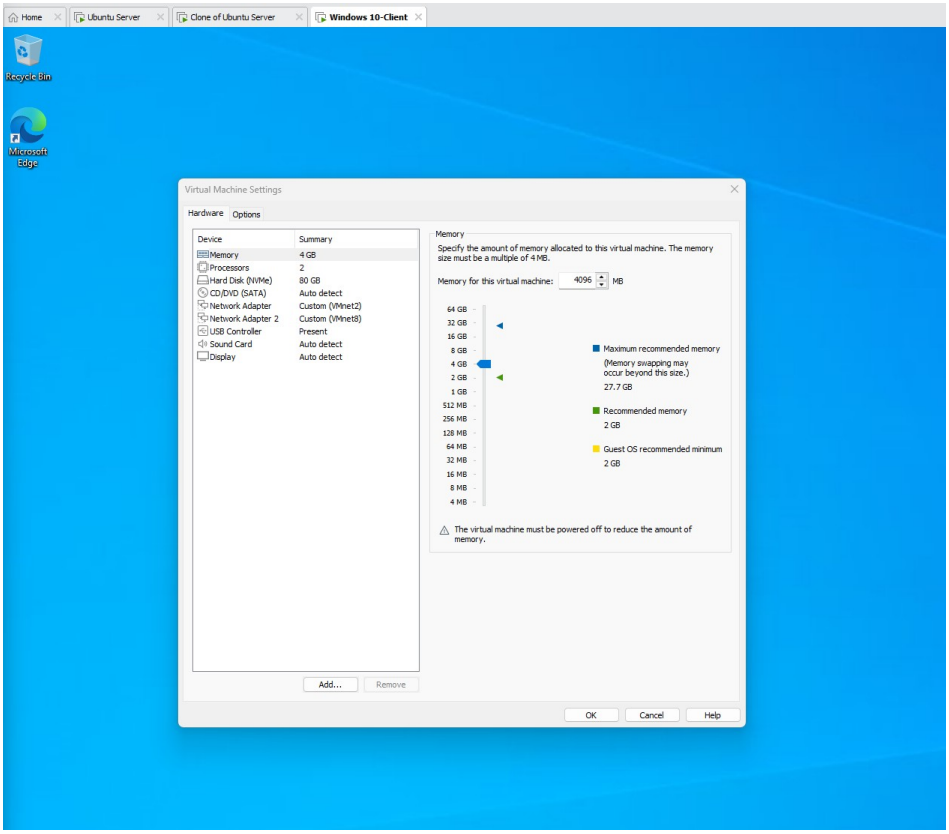


Figure 3. *win-client VM settings: 4 GB RAM, 2 processors, 80 GB NVMe disk (setup stage)*

> **What this proves:** the three VMs exist with the roles and resources in the machine table (L1).

L2 Lab network and static IPs

Each VM has a static IP on VMnet2: siem-host 10.10.10.10, ubuntu-web 10.10.10.50, win-client 10.10.10.80. siem-host keeps a permanent NAT adapter to pull Docker images. The endpoints had NAT only while installing software, and it was removed afterwards (see S2).

L3 Connectivity, dashboard and clocks

All three VMs can ping each other, and the Dell can ping all three.

From \ To	siem-host	ubuntu-web	win-client
siem-host	—	OK	OK
ubuntu-web	OK	—	OK
win-client	OK	OK	—
Dell (host)	OK	OK	OK

```
ubuntu@siem-host:~$ ping 10.10.10.50
PING 10.10.10.50 (10.10.10.50) 56(84) bytes of data.
64 bytes from 10.10.10.50: icmp_seq=1 ttl=64 time=0.298 ms
64 bytes from 10.10.10.50: icmp_seq=2 ttl=64 time=1.02 ms
64 bytes from 10.10.10.50: icmp_seq=3 ttl=64 time=0.777 ms
64 bytes from 10.10.10.50: icmp_seq=4 ttl=64 time=0.217 ms

^C--- 10.10.10.50 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3045ms
rtt min/avg/max/mdev = 0.217/0.579/1.024/0.334 ms
ubuntu@siem-host:~$ ping 10.10.10.80
PING 10.10.10.80 (10.10.10.80) 56(84) bytes of data.
64 bytes from 10.10.10.80: icmp_seq=1 ttl=128 time=0.499 ms
64 bytes from 10.10.10.80: icmp_seq=2 ttl=128 time=0.922 ms
64 bytes from 10.10.10.80: icmp_seq=3 ttl=128 time=1.62 ms
^C
--- 10.10.10.80 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 0.499/1.015/1.624/0.463 ms
ubuntu@siem-host:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:24:55:ed brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    altname enx000c292455ed
    inet 10.10.10.10/24 brd 10.10.10.255 scope global ens37
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe24:55ed/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
3: ens38: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:24:55:f7 brd ff:ff:ff:ff:ff:ff
    altname enp2s6
    altname enx000c292455f7
    inet 192.168.121.156/24 metric 100 brd 192.168.121.255 scope global dynamic ens38
       valid_lft 972sec preferred_lft 972sec
    inet6 fe80::20c:29ff:fe24:55f7/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
ubuntu@siem-host:~$
```

Figure 4. *siem-host: ip a (10.10.10.10) and ping to ubuntu-web and win-client*

```
ubuntu@ubuntu-web:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:83:e6:d4 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c2983e6d4
    inet 10.10.10.50/24 brd 10.10.10.255 scope global ens33
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe83:e6d4/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
ubuntu@ubuntu-web:~$ ping 10.10.10.80
PING 10.10.10.80 (10.10.10.80) 56(84) bytes of data.
64 bytes from 10.10.10.80: icmp_seq=1 ttl=128 time=0.263 ms
64 bytes from 10.10.10.80: icmp_seq=2 ttl=128 time=0.773 ms
^C
--- 10.10.10.80 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1045ms
rtt min/avg/max/mdev = 0.263/0.518/0.773/0.255 ms
ubuntu@ubuntu-web:~$ ping 10.10.10.10
PING 10.10.10.10 (10.10.10.10) 56(84) bytes of data.
64 bytes from 10.10.10.10: icmp_seq=1 ttl=64 time=0.373 ms
64 bytes from 10.10.10.10: icmp_seq=2 ttl=64 time=0.400 ms
64 bytes from 10.10.10.10: icmp_seq=3 ttl=64 time=0.987 ms
^C
--- 10.10.10.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 1999ms
rtt min/avg/max/mdev = 0.373/0.586/0.987/0.283 ms
ubuntu@ubuntu-web:~$ _
```

Figure 5. *ubuntu-web: ip a (only ens33, 10.10.10.50) and ping to win-client and siem-host*

```
C:\Users\Student>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::d5be:466:5c04:8769%4
   IPv4 Address. . . . . . . . . . . : 10.10.10.80
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 10.10.10.1

C:\Users\Student>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:
Reply from 10.10.10.10: bytes=32 time<1ms TTL=64
Reply from 10.10.10.10: bytes=32 time=1ms TTL=64

Ping statistics for 10.10.10.10:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
Control-C
^C
C:\Users\Student>ping 10.10.10.50

Pinging 10.10.10.50 with 32 bytes of data:
Reply from 10.10.10.50: bytes=32 time<1ms TTL=64
Reply from 10.10.10.50: bytes=32 time<1ms TTL=64

Ping statistics for 10.10.10.50:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
Control-C
^C
C:\Users\Student>
```

Figure 6. *win-client: ipconfig (10.10.10.80) and ping to siem-host and ubuntu-web*

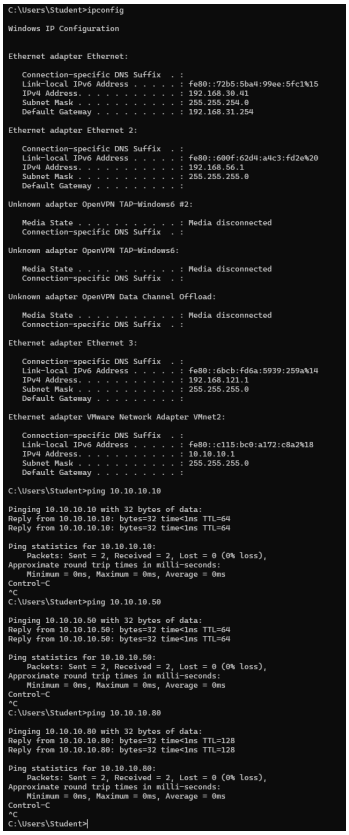


Figure 7. *Dell: VMnet2 adapter 10.10.10.1, successful ping to all three VMs*

All three machines use the **Asia/Baku (UTC+4)** time zone. NTP is active on the Linux VMs. On Windows I fixed the time zone with `Set-TimeZone`. This matters so that Wazuh alerts are ordered correctly in time. The dashboard opens from the Dell's browser at `https://10.10.10.10` (see W2).

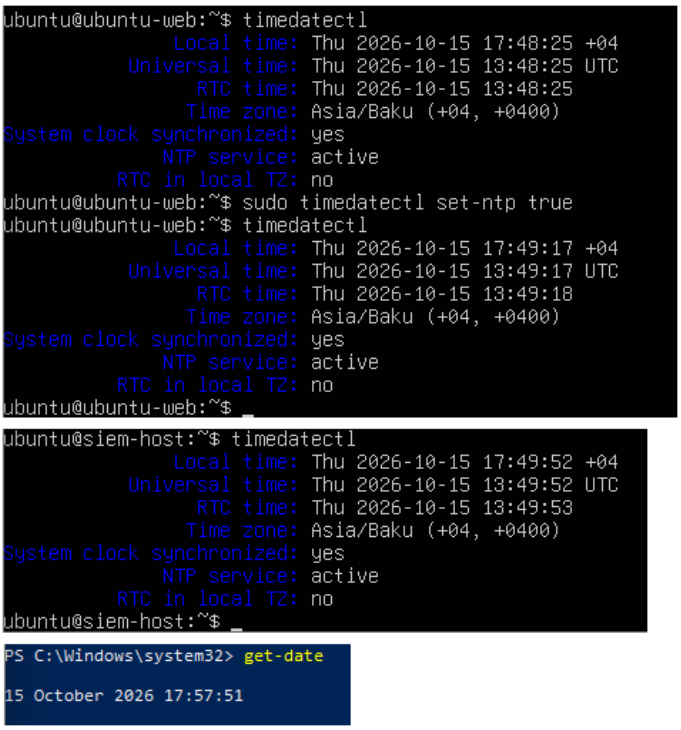


Figure 8. *Clocks: timedatectl on ubuntu-web and siem-host (synchronized: yes), Get-Date on win-client*

L4 Docker and the kernel setting

Docker Engine and the Compose plugin were installed on siem-host from Docker's official apt repository. The `vm.max_map_count = 262144` setting required by the Wazuh indexer is written to `/etc/sysctl.d/99-wazuh.conf`, so it survives a reboot (see Figure 16).

```
ubuntu@siem-host:~$ docker --version
Docker version 29.9.1, build dfc520d
ubuntu@siem-host:~$ docker compose version
Docker Compose version v5.6.0
ubuntu@siem-host:~$ sysctl vm.max_map_count
vm.max_map_count = 262144
ubuntu@siem-host:~$ sudo docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

ubuntu@siem-host:~$
```

Figure 9. *Docker 29.9.1, Compose v5.6.0, vm.max_map_count = 262144 and the hello-world test*

> **Problem and fix:** Pulling the images failed with “no space left on device”. The disk was 256 GB, but the Ubuntu installer had made the LVM volume only ~18 GB. I grew it to ~251 GB with `growpart`, `pvresize` and `lvextend -r`.

L5 Snapshots

On all three VMs (siem-host, ubuntu-web, win-client) I took a snapshot named `w1-clean-install` before starting any Wazuh work, and one named `w1-done` at the end of the week. I took them from VMware → VM → Snapshot → Take Snapshot. Any VM can now be rolled back to either state.

3 Wazuh platform W1–W5

The Wazuh **4.14.7** single-node stack runs in Docker on siem-host. The dashboard opens over HTTPS, the default password has been changed, and data survives both a stop/start and a reboot.

W1 Bringing up the stack

The stack was started from the `v4.14.7` tag of the wazuh-docker repository with `docker compose up -d`. Three containers run: manager, indexer and dashboard. All three images are the same version.

```
ubuntu@siem-host:~/wazuh-docker/single-node$ docker compose ps
NAME                          IMAGE                          COMMAND                  SERVICE           CREATED         STATUS
     PORTS
single-node-wazuh.dashboard-1   wazuh/wazuh-dashboard:4.14.7   "/entrypoint.sh"         wazuh.dashboard   7 minutes ago   Up 7 minut
es   443/tcp, 0.0.0.0:443->5601/tcp, [::]:443->5601/tcp
single-node-wazuh.indexer-1     wazuh/wazuh-indexer:4.14.7     "/entrypoint.sh open…"   wazuh.indexer     7 minutes ago   Up 7 minut
es   0.0.0.0:9200->9200/tcp, [::]:9200->9200/tcp
single-node-wazuh.manager-1     wazuh/wazuh-manager:4.14.7     "/init"                  wazuh.manager     7 minutes ago   Up 7 minut
es   0.0.0.0:1514-1515->1514-1515/tcp, [::]:1514-1515->1514-1515/tcp, 0.0.0.0:514->514/udp, [::]:514->514/udp, 0.0.0.0:55000->55000/
tcp, [::]:55000->55000/tcp, 1516/tcp
ubuntu@siem-host:~/wazuh-docker/single-node$ df -h /
Filesystem                         Size  Used Avail Use% Mounted on
/dev/mapper/ubuntu--vg-ubuntu--lv  251G   19G  222G   8% /
ubuntu@siem-host:~/wazuh-docker/single-node$
```

Figure 10. *docker compose ps: three containers Up, images 4.14.7; df -h: root volume 251 GB*

W2 HTTPS dashboard and the certificate warning

The dashboard opens over HTTPS from the Dell's browser. The browser shows “Your connection is not private” (NET::ERR_CERT_AUTHORITY_INVALID). The reason: the certificate was generated by Wazuh's own tool (**self-signed**) and is not signed by a trusted certificate authority (CA). Traffic is still encrypted; the browser simply cannot verify the server's identity. This is normal for a lab.

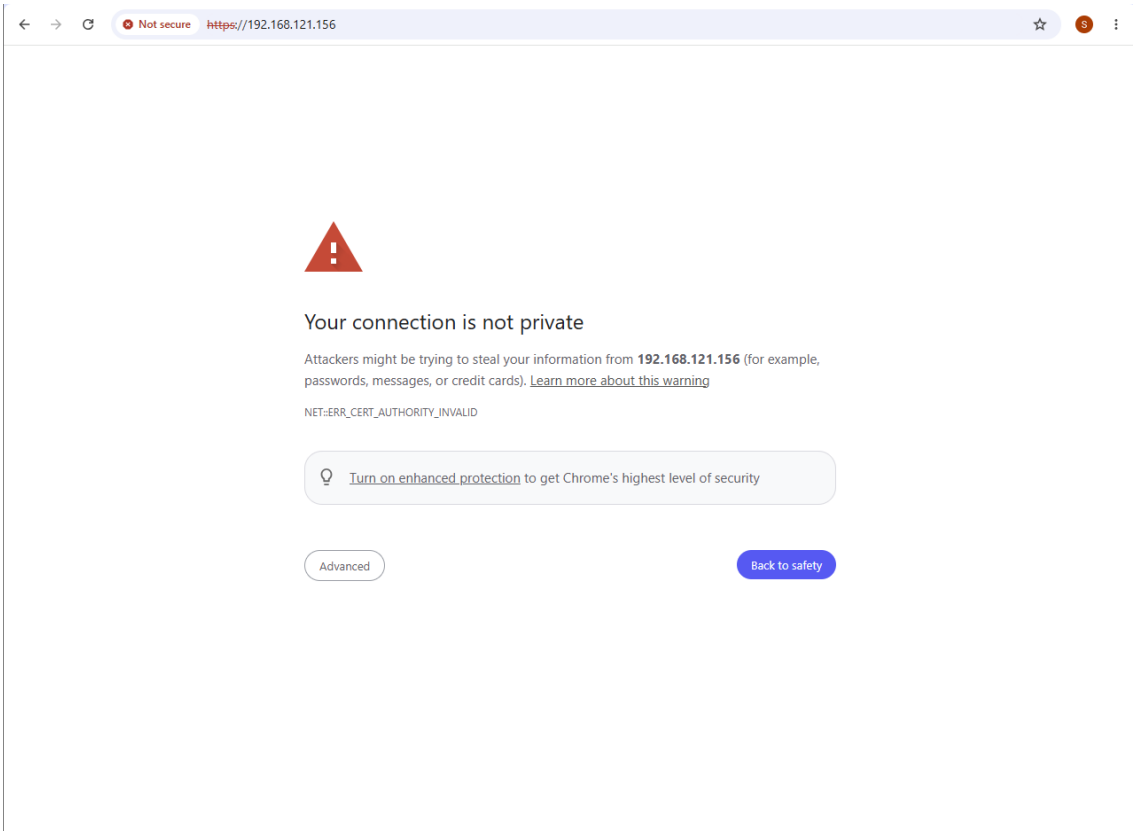


Figure 11. *Certificate warning; taken via the NAT address, the certificate is the same on the lab address*

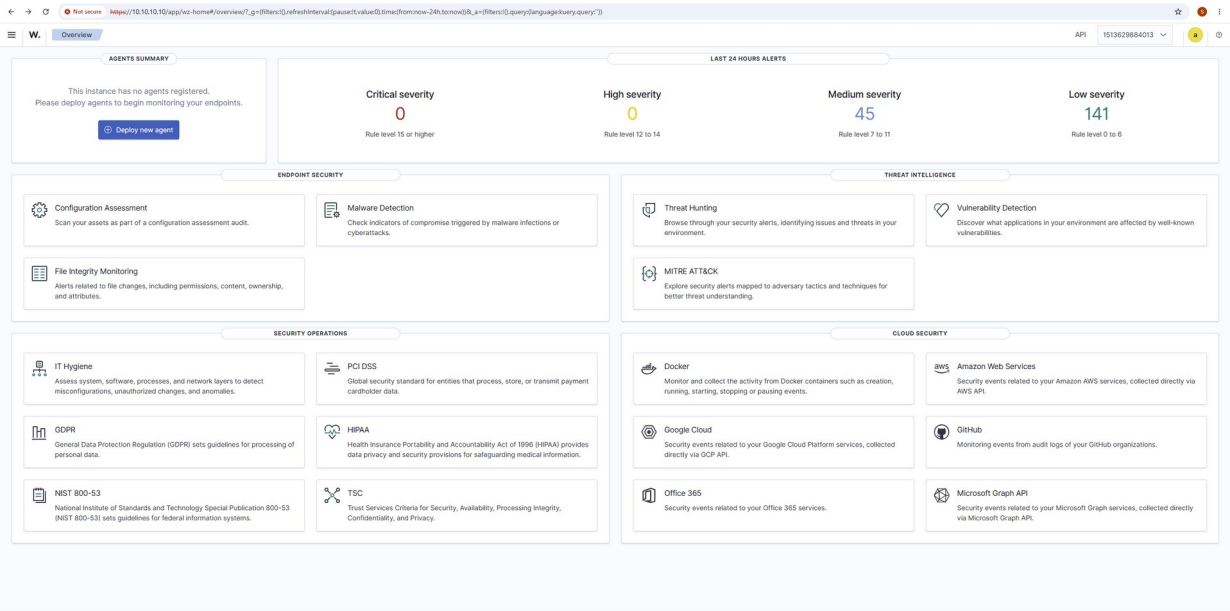


Figure 12. *Dashboard home page after logging in (https://10.10.10.10)*

W3 Changing the default password

I changed the default admin password using Wazuh's documented procedure for the Docker deployment:

1. Stopped the stack and generated a hash of the new password with the `wazuh-indexer` image.
2. Wrote the hash for the admin user in `config/wazuh_indexer/internal_users.yml`.
3. Updated the `INDEXER_PASSWORD` value in `docker-compose.yml`.
4. Started the stack and applied the change with `securityadmin.sh` inside the indexer container.

The new password is not stored in Git or in this report.

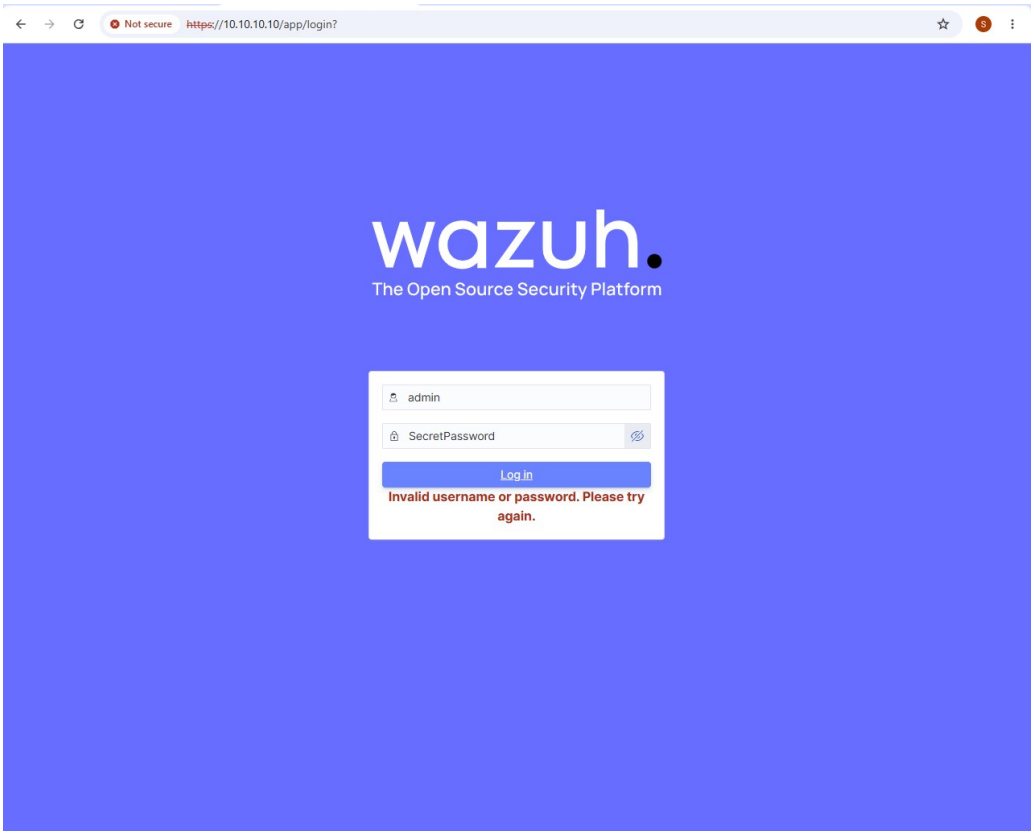


Figure 13. *The old default password is now rejected: "Invalid username or password"*

> **Problem and fix:** When the password differed between the two files, the dashboard returned a 401 error. Writing the same password in both files fixed it.

W4 Persistence: stop/start and reboot

`docker compose stop` and `docker compose start` lost no data. I did not use `docker compose down -v`, because `-v` deletes the data volumes.

```
ubuntu@siem-host:~/wazuh-docker/single-node$ docker compose stop
[+] stop 3/3
 ✔Container single-node-wazuh.dashboard-1 Stopped
 ✔Container single-node-wazuh.manager-1   Stopped
 ✔Container single-node-wazuh.indexer-1   Stopped
ubuntu@siem-host:~/wazuh-docker/single-node$ docker compose start
[+] start 3/3
 ✔Container single-node-wazuh.manager-1   Started
 ✔Container single-node-wazuh.indexer-1   Started
 ✔Container single-node-wazuh.dashboard-1 Started
```

Figure 14. *docker compose stop and start*

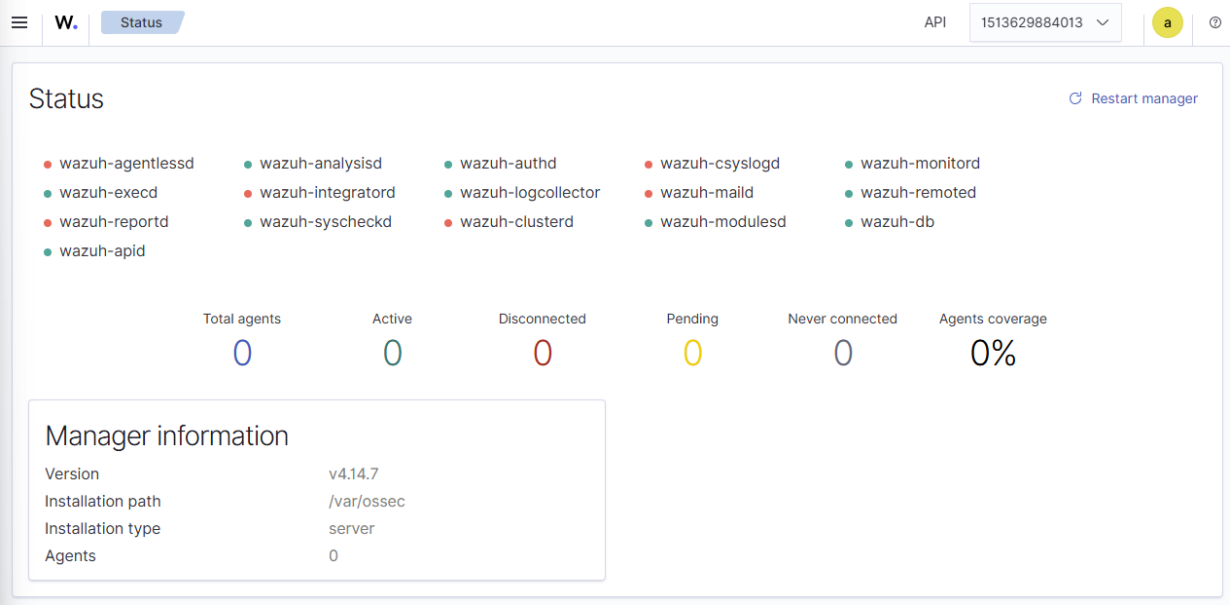


Figure 15. *After stop/start the manager runs (v4.14.7); the red daemons are modules not used in this setup*

I then rebooted siem-host. The containers **came back up on their own** thanks to the `restart: always` policy. `vm.max_map_count` and the IP addresses were also preserved. Nothing had to be done by hand.

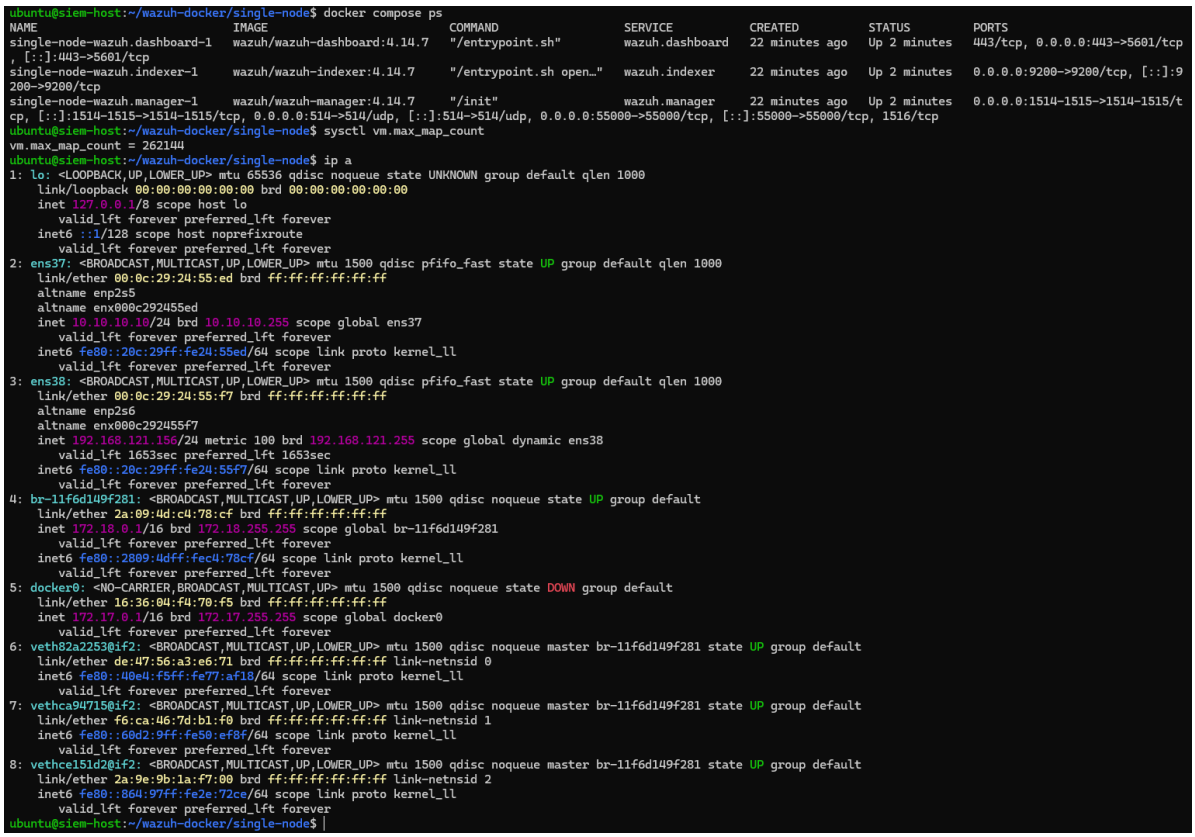


Figure 16. *After reboot: containers Up, vm.max_map_count = 262144, IPs in place*

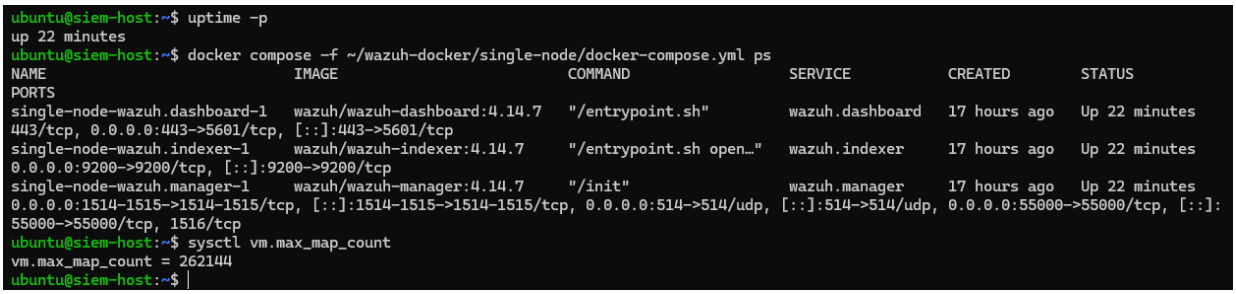


Figure 17. *Reboot proof: uptime 22 minutes; containers created 17 hours ago, Up for 22 minutes*

I repeated the test after the agents were enrolled. After the reboot both agents became **Active** again and the earlier alerts were still in the dashboard.

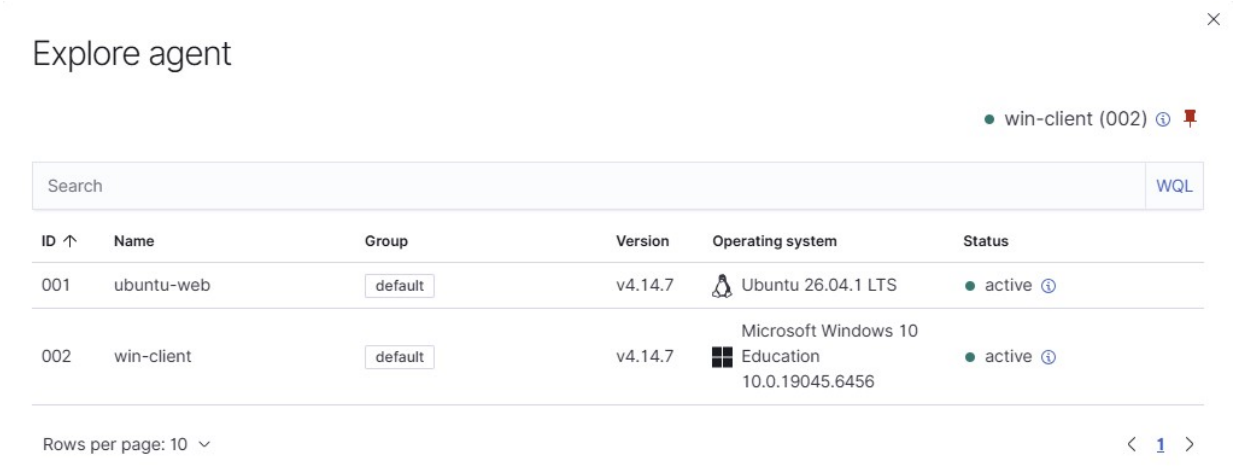


Figure 18. *After reboot: ubuntu-web (001) and win-client (002) Active — "still alive after reboot"*

W5 Container memory usage

I measured each container's memory twice with `docker stats --no-stream`: right after start-up and once the containers had settled.

Container	1st measurement (start-up)	2nd measurement (idle)
wazuh-indexer	1.522 GiB	1.454 GiB
wazuh-manager	702.4 MiB	402.2 MiB
wazuh-dashboard	207.8 MiB	186.5 MiB
Total	**~2.4 GiB**	**~2.0 GiB**

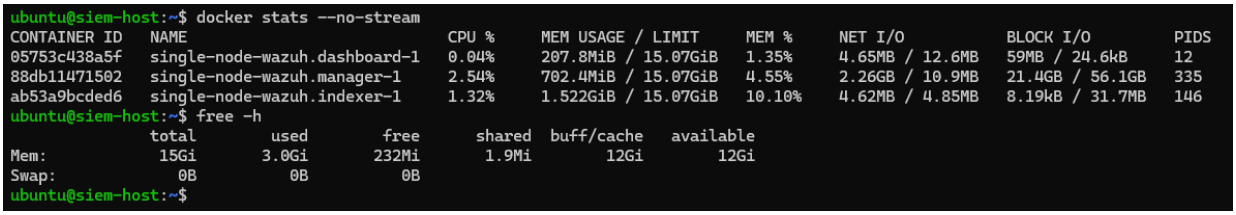


Figure 19. *1st measurement: docker stats and free -h*

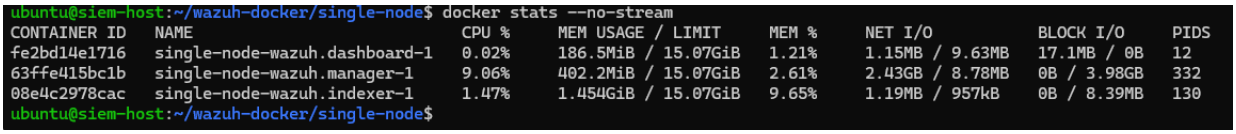


Figure 20. *2nd measurement: idle state*

The indexer uses the most memory because it is built on OpenSearch and reserves a JVM heap.

4 Agents and telemetry A1–A4

Wazuh agent **4.14.7** runs on both endpoints and shows as **Active** in the dashboard. Sysmon events arrive from win-client, and Apache runs on ubuntu-web.

A1 ubuntu-web agent

I installed the agent from the `.deb` package with manager address `WAZUH_MANAGER=10.10.10.10` and agent name `WAZUH_AGENT_NAME=ubuntu-web`, then enabled it with `systemctl enable` and `start`.

```
ubuntu@ubuntu-web:~$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.7-1_amd64.deb && sudo WAZUH_MANAGER=
'10.10.10.10' WAZUH_AGENT_NAME='ubuntu-web' dpkg -i ./wazuh-agent_4.14.7-1_amd64.deb
--2026-10-16 13:36:04--  https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.7-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 65.9.9.50, 65.9.9.98, 65.9.9.45, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|65.9.9.50|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13222832 (13M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.14.7-1_amd64.deb'

wazuh-agent_4.14.7-1_amd64.deb     100%[===================================================>]  12.61M  12.2MB/s    in 1.0s

2026-10-16 13:36:06 (12.2 MB/s) - 'wazuh-agent_4.14.7-1_amd64.deb' saved [13222832/13222832]

[sudo: authenticate] Password:
Selecting previously unselected package wazuh-agent.
(Reading database… 98687 files and directories currently installed.)
Preparing to unpack …/wazuh-agent_4.14.7-1_amd64.deb…
Unpacking wazuh-agent (4.14.7-1)…
Setting up wazuh-agent (4.14.7-1)…
ubuntu@ubuntu-web:~$ sudo systemctl daemon-reload
ubuntu@ubuntu-web:~$ sudo systemctl enable wazuh-agent
Created symlink '/etc/systemd/system/multi-user.target.wants/wazuh-agent.service' → '/usr/lib/systemd/system/wazuh-agent.service'.
ubuntu@ubuntu-web:~$ sudo systemctl start wazuh-agent
ubuntu@ubuntu-web:~$ sudo systemctl status wazuh-agent --no-pager
● wazuh-agent.service - Wazuh agent
     Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-10-16 13:37:04 +04; 9s ago
 Invocation: 652351ddff014c9b9898bcd5261e1dde
    Process: 4762 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
      Tasks: 29 (limit: 1824)
     Memory: 24.8M (peak: 29.1M)
        CPU: 352ms
     CGroup: /system.slice/wazuh-agent.service
             ├─4784 /var/ossec/bin/wazuh-execd
             ├─4793 /var/ossec/bin/wazuh-agentd
             ├─4805 /var/ossec/bin/wazuh-syscheckd
             ├─4816 /var/ossec/bin/wazuh-logcollector
             └─4833 /var/ossec/bin/wazuh-modulesd

Oct 16 13:36:59 ubuntu-web systemd[1]: Starting wazuh-agent.service - Wazuh agent...
Oct 16 13:36:59 ubuntu-web env[4762]: Starting Wazuh v4.14.7...
Oct 16 13:36:59 ubuntu-web env[4762]: Started wazuh-execd...
Oct 16 13:37:00 ubuntu-web env[4762]: Started wazuh-agentd...
Oct 16 13:37:00 ubuntu-web env[4762]: Started wazuh-syscheckd...
Oct 16 13:37:01 ubuntu-web env[4762]: Started wazuh-logcollector...
Oct 16 13:37:02 ubuntu-web env[4762]: Started wazuh-modulesd...
Oct 16 13:37:04 ubuntu-web env[4762]: Completed.
Oct 16 13:37:04 ubuntu-web systemd[1]: Started wazuh-agent.service - Wazuh agent.
ubuntu@ubuntu-web:~$
```

Figure 21. *ubuntu-web: wazuh-agent 4.14.7 installation and systemctl status: active (running)*

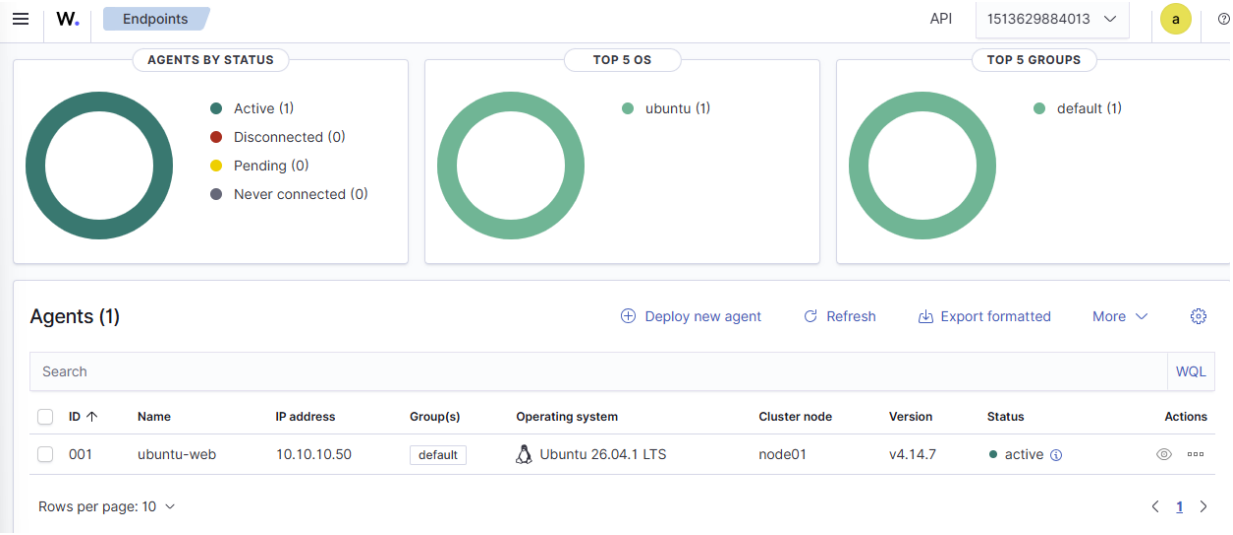


Figure 22. *Dashboard → Endpoints: ubuntu-web (001), 10.10.10.50, v4.14.7, active*

A2 win-client agent

I installed the agent in PowerShell from the `.msi` package with `WAZUH_MANAGER='10.10.10.10'` and `WAZUH_AGENT_NAME='win-client'`. The WazuhSvc service is Running.

> **Problem and fix:** On the first attempt I mistakenly entered win-client's own IP instead of the manager address, and the agent did not connect. After changing it to 10.10.10.10 the agent connected.

```
PS C:\Windows\system32> Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.7-1.msi -OutFile $env:tmp\wazuh-agent;
 msiexec.exe /i $env:tmp\wazuh-agent /q WAZUH_MANAGER='10.10.10.10' WAZUH_AGENT_NAME='win-client'
PS C:\Windows\system32> NET START Wazuh
The requested service has already been started.

More help is available by typing NET HELPMSG 2182.

PS C:\Windows\system32> Get-Service Wazuh

Status   Name               DisplayName
------   ----               -----------
Running  WazuhSvc           Wazuh
```

Figure 23. *win-client: agent installation command and WazuhSvc: Running*

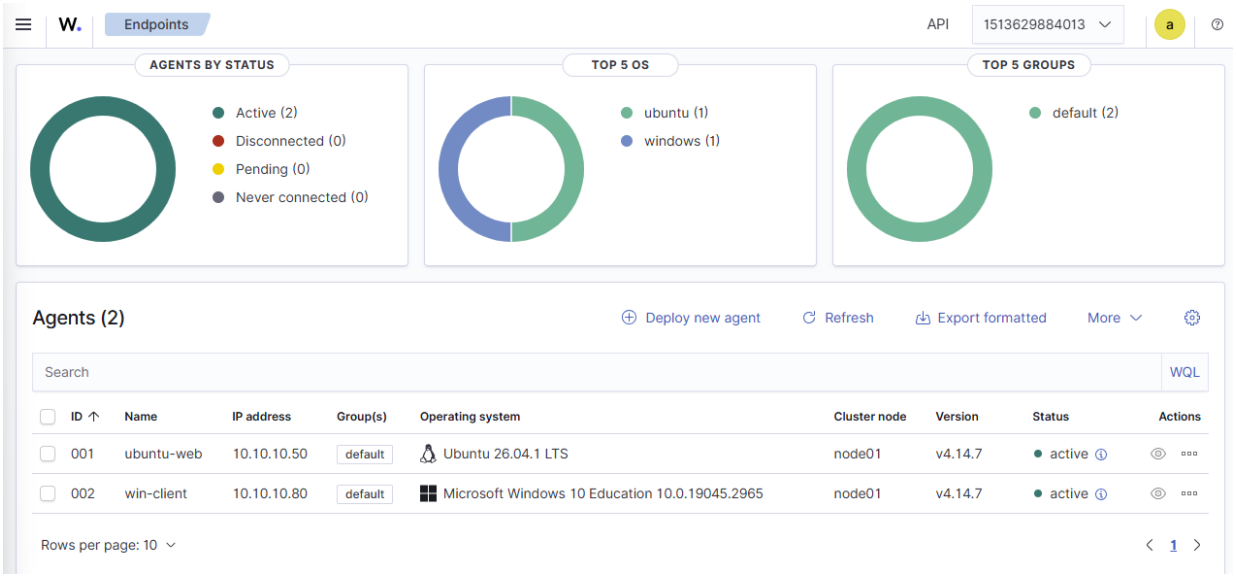


Figure 24. *Dashboard: two agents Active — ubuntu-web (001) and win-client (002)*

A3 Sysmon

I installed Sysinternals Sysmon on win-client with SwiftOnSecurity's `sysmonconfig-export.xml` community configuration. I then added this block to the agent's `ossec.conf` and restarted the WazuhSvc service:

```
<localfile>
  <location>Microsoft-Windows-Sysmon/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
```

```
PS C:\Windows\system32> cd $env:USERPROFILE\Downloads
PS C:\Users\Student\Downloads> Invoke-WebRequest -Uri "https://download.sysinternals.com/files/Sysmon.zip" -OutFile Sysmon.zip
PS C:\Users\Student\Downloads> Expand-Archive Sysmon.zip -DestinationPath Sysmon -Force
PS C:\Users\Student\Downloads> Invoke-WebRequest -Uri "https://raw.githubusercontent.com/SwiftOnSecurity/sysmon-config/master/sysmonconfig-ex
port.xml" -OutFile sysmonconfig-export.xml
PS C:\Users\Student\Downloads> cd Sysmon
PS C:\Users\Student\Downloads\Sysmon> .\Sysmon64.exe -accepteula -i ..\sysmonconfig-export.xml


System Monitor v15.22 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.91
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
PS C:\Users\Student\Downloads\Sysmon> Get-Service Sysmon64

Status   Name               DisplayName
------   ----               -----------
Running  Sysmon64           Sysmon64


PS C:\Users\Student\Downloads\Sysmon> notepad "C:\Program Files (x86)\ossec-agent\ossec.conf"
PS C:\Users\Student\Downloads\Sysmon> Restart-Service WazuhSvc
PS C:\Users\Student\Downloads\Sysmon>
```

Figure 25. *Sysmon64 installed (Running), ossec.conf edited and Restart-Service WazuhSvc*

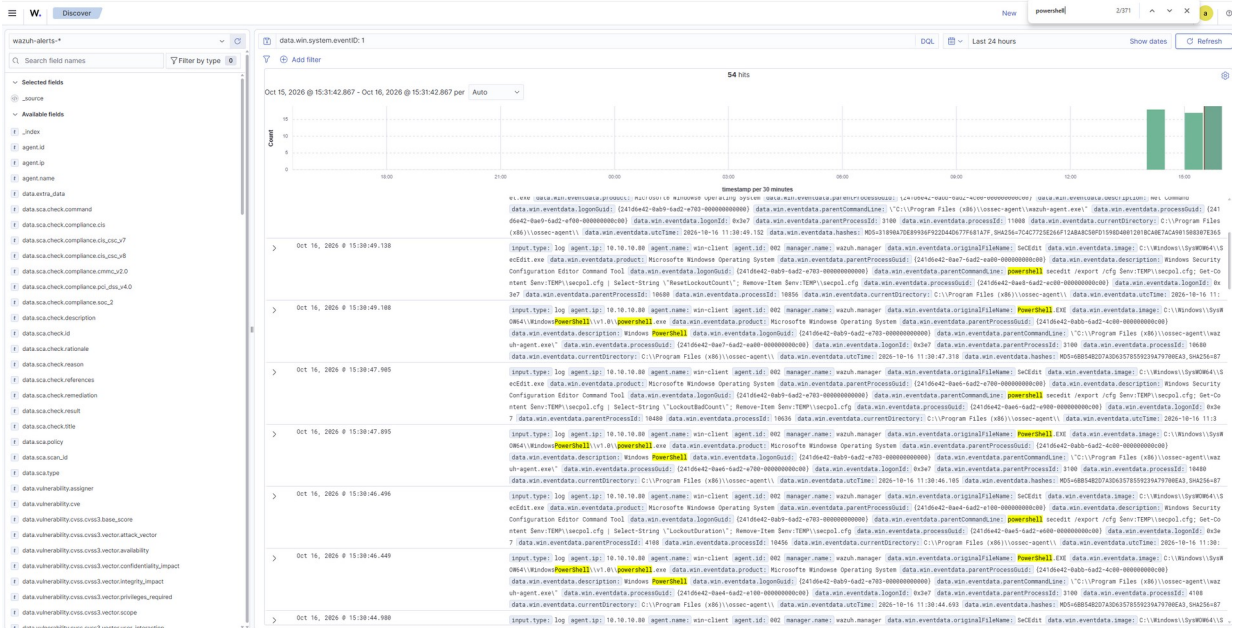


Figure 26. *Discover: data.win.system.eventID: 1 (Process Create), agent.name: win-client*

> **What does Sysmon add?** The standard Windows Security log can record that a process started. Sysmon also records the full command line, the parent process, the file hash, network connections and DNS queries. Some of the events in the screenshot are generated by the agent itself (parent `wazuh-agent.exe`); this is "background noise" to be filtered in later weeks.

A4 Web server

Apache2 and **SSH** run on ubuntu-web. Apache's default page opens from the Dell's browser. This server will be the attack target in Week 3.

```
ubuntu@ubuntu-web:~$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-10-16 13:39:48 +04; 8s ago
 Invocation: da6250867ccb4fbebf8ad9f6e779c3b9
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 6695 (apache2)
     Status: "Processing requests..."
      Tasks: 55 (limit: 1824)
     Memory: 5.3M (peak: 5.6M)
        CPU: 37ms
     CGroup: /system.slice/apache2.service
             ├─6695 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─6711 /usr/sbin/apache2 -k start -DFOREGROUND
             └─6712 /usr/sbin/apache2 -k start -DFOREGROUND

Oct 16 13:39:48 ubuntu-web systemd[1]: Starting apache2.service - The Apache HTTP Server...
Oct 16 13:39:48 ubuntu-web apachectl[6695]: AH00558: apache2: Could not reliably determine the server's fully qualified domain n…s message
Oct 16 13:39:48 ubuntu-web systemd[1]: Started apache2.service - The Apache HTTP Server.
Hint: Some lines were ellipsized, use -l to show in full.
ubuntu@ubuntu-web:~$ sudo systemctl status ssh --no-pager
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
     Active: active (running) since Fri 2026-10-16 13:31:22 +04; 8min ago
 Invocation: 0ac72ecb3ea04f0885073a931f3042a5
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 2671 (sshd)
      Tasks: 1 (limit: 1824)
     Memory: 3.3M (peak: 5.4M)
        CPU: 41ms
     CGroup: /system.slice/ssh.service
             └─2671 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Oct 16 13:31:22 ubuntu-web systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Oct 16 13:31:22 ubuntu-web sshd[2671]: Server listening on 0.0.0.0 port 22.
Oct 16 13:31:22 ubuntu-web sshd[2671]: Server listening on :: port 22.
Oct 16 13:31:22 ubuntu-web systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Oct 16 13:31:27 ubuntu-web sshd-session[2674]: Accepted password for ubuntu from 10.10.10.1 port 54385 ssh2
Oct 16 13:31:27 ubuntu-web sshd-session[2674]: pam_unix(sshd:session): session opened for user ubuntu(uid=1000) by ubuntu(uid=0)
ubuntu@ubuntu-web:~$
```

Figure 27. *ubuntu-web: apache2 and ssh services active (running)*



Figure 28. *http://10.10.10.50 in the Dell's browser: Apache2 Ubuntu Default Page*

5 Verification V1–V2

I deliberately caused five events and found all five in the dashboard. Repeated wrong passwords additionally raised a level 10 alert.

V1 Five known events

#	Event	Rule description	Level	Rule ID	Evidence
1	Failed SSH login (ubuntu-web)	sshd: authentication failed.	5	5760	Figure 29
2	Command run with sudo (ubuntu-web)	Successful sudo to ROOT executed.	3	5402	Figure 30
3	New local user (ubuntu-web)	New user added to the system.	8	5902	Figure 30
4	Program started on Windows (Sysmon)	Discovery activity executed (Sysmon Event ID 1)	3	92031	Figures 31, 32
5	Failed Windows logon (win-client)	Logon Failure - Unknown user or bad password	5	60122	Figures 33, 34

1. Failed SSH login. From the Dell I ran `ssh ubuntu@10.10.10.50` and typed a wrong password several times, and also tried a non-existent user. Wazuh fired rules 5760 (authentication failed), 5557 (password check failed), 5503 (login failed) and 5710 (non-existent user).

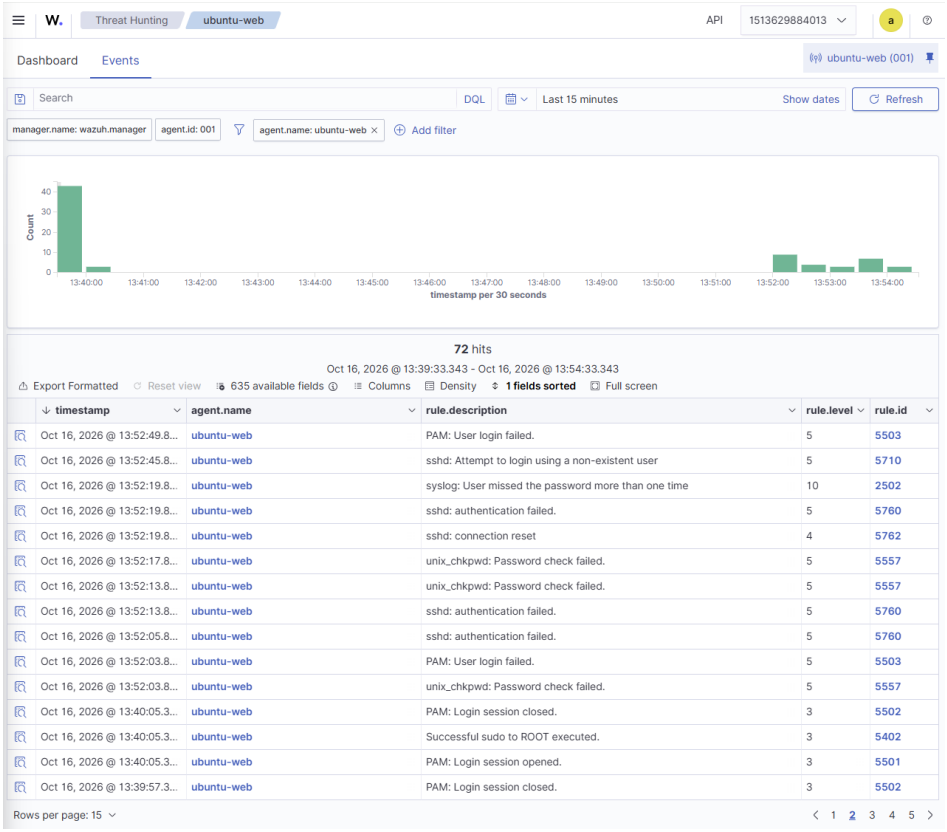


Figure 29. *ubuntu-web: failed SSH login (5760, 5557, 5503), non-existent user (5710) and repeated wrong password (2502, level 10)*

2–3. sudo and a new user. I ran a command with `sudo` (5402), then created a new local user with `sudo adduser`. This produced alerts 5902 (new user), 5901 (new group), 5904 (user information changed) and 5555 (password changed).

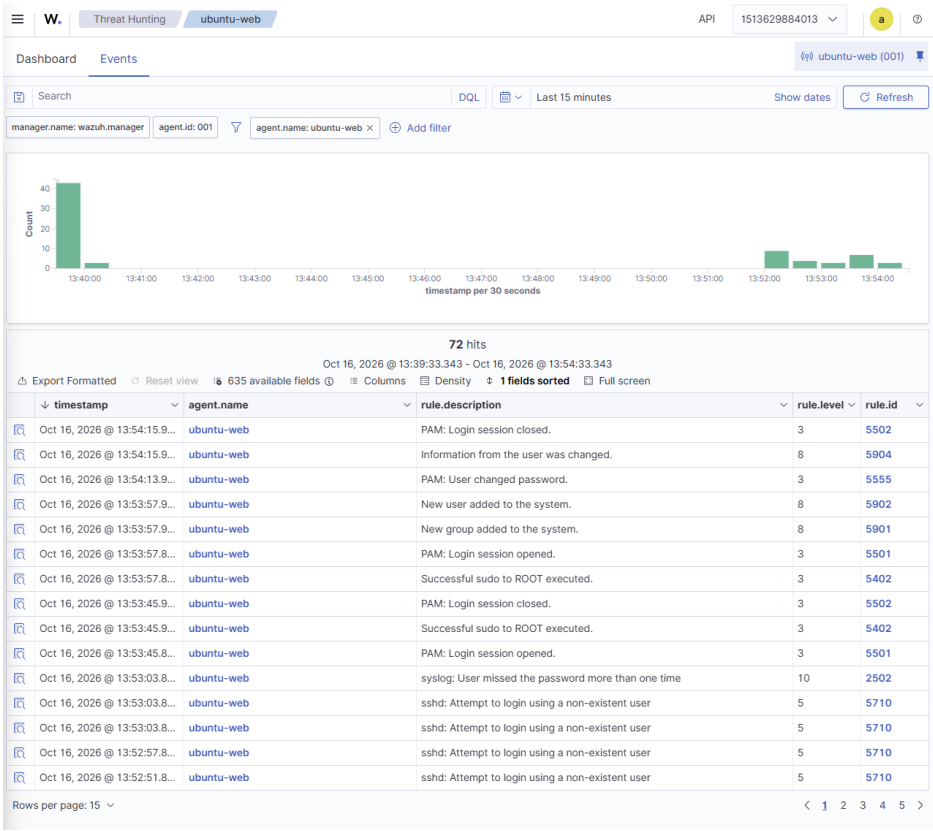


Figure 30. *ubuntu-web: sudo (5402) and new user creation (5902, 5901, 5904, 5555)*

4. Program started on Windows (Sysmon). On win-client I ran `net user` in cmd. Sysmon recorded it as Process Create (Event ID 1): `net.exe` internally starts `net1.exe` (`commandLine: net1 user`, `parentImage: net.exe`, user Student). Wazuh turned it into an alert with rule **92031 · Discovery activity executed**, **level 3**, mapped to MITRE ATT&CK T1087 (Account Discovery).

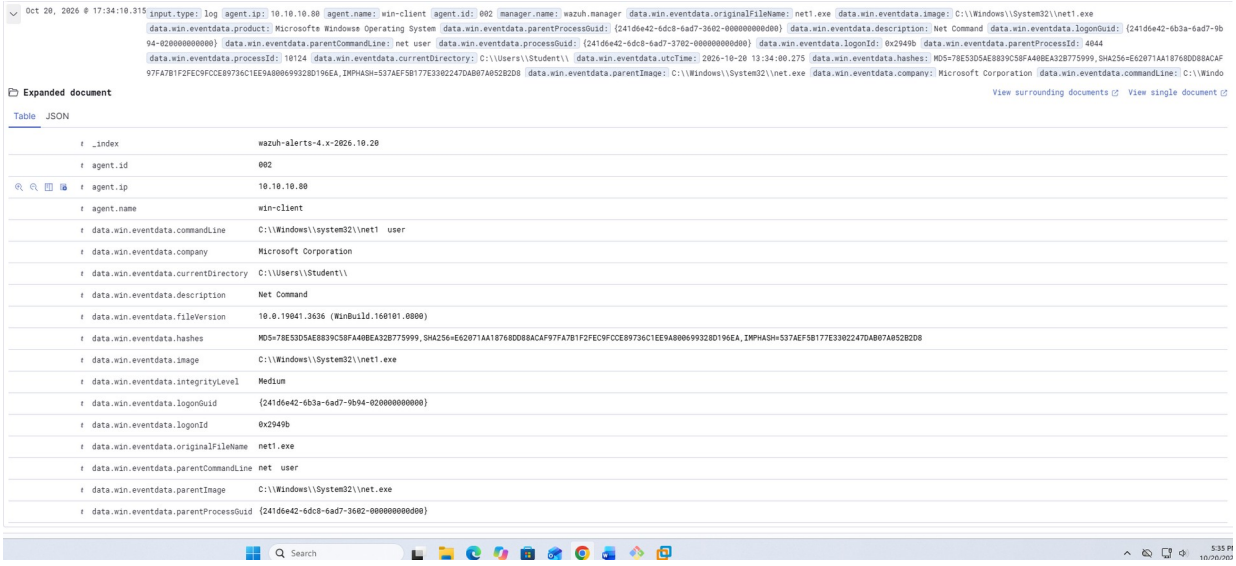


Figure 31. *Sysmon Event ID 1: the net user command — image net1.exe, parentImage net.exe, commandLine "net1 user", agent win-client*

t	data.win.system.processID	2996
t	data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}
t	data.win.system.providerName	Microsoft-Windows-Sysmon
t	data.win.system.severityValue	INFORMATION
t	data.win.system.systemTime	2026-10-20T13:34:00.2782961Z
t	data.win.system.task	1
t	data.win.system.threadID	4836
t	data.win.system.version	5
t	decoder.name	windows_eventchannel
t	id	1792503250.404378
t	input.type	log
t	location	EventChannel
t	manager.name	wazuh.manager
t	rule.description	Discovery activity executed
#	rule.firedtimes	8
t	rule.groups	sysmon, sysmon_eid1_detections, windows
t	rule.id	92031
#	rule.level	3
◕	rule.mail	false
t	rule.mitre.id	T1087
t	rule.mitre.tactic	Discovery
t	rule.mitre.technique	Account Discovery
🗓	timestamp	Oct 20, 2026 @ 17:34:10.315

Figure 32. *Rule fields of the same alert: rule.id 92031, rule.level 3, "Discovery activity executed", group sysmon_eid1_detections, MITRE T1087*

5. Failed Windows logon. On win-client I attempted to log in as a non-existent user `yoxdur` with a wrong password. Windows logged it in the Security log as Event ID 4625 (`subStatus 0xc0000064` — no such user, `logonType 2` — interactive logon). Wazuh raised a level 5 alert with rule 60122.

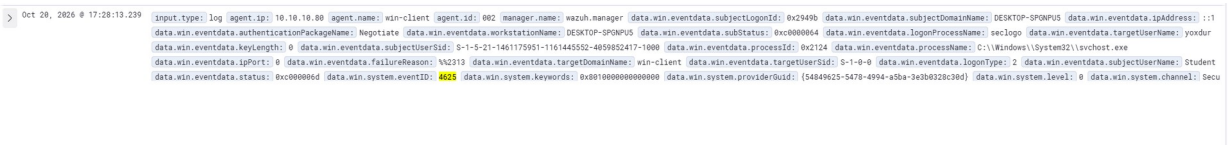


Figure 33. *win-client: failed Windows logon — Event ID 4625, targetUserName: yoxdur, logonType 2, subStatus 0xc0000064*

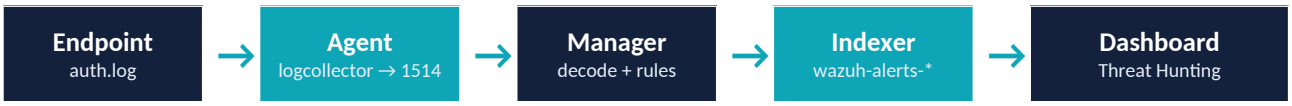
data.win.system.task	12544
data.win.system.threadID	744
data.win.system.version	0
decoder.name	windows_eventchannel
id	1792502893.374118
input.type	log
location	EventChannel
manager.name	wazuh.manager
rule.description	Logon Failure - Unknown user or bad password
rule.firedtimes	1
rule.gdpr	IV_35.7.d, IV_32.2
rule.gpg13	7.1
rule.groups	windows, windows_security, authentication_failed
rule.hipaa	164.312.b
rule.id	60122
rule.level	5
rule.mail	false
rule.mitre.id	T1531
rule.mitre.tactic	Impact
rule.mitre.technique	Account Access Removal
rule.nist_800_53	AU.14, AC.7
rule.pci_dss	10.2.4, 10.2.5
rule.tsc	CC6.1, CC6.8, CC7.2, CC7.3
timestamp	Oct 20, 2026 @ 17:28:13.239

Figure 34. *Rule fields of the same alert: "Logon Failure - Unknown user or bad password", rule.level 5, rule.id 60122, MITRE T1531*

> **Observation:** Repeated wrong-password attempts are treated as more serious than a single one: "User missed the password more than one time" (rule 2502) got **level 10**, while a single failed login is level 5.

V2 Path of an event from endpoint to dashboard

Example: an SSH attempt with a wrong password from the Dell to ubuntu-web.



1. **The event happens.** sshd writes the failed login to `/var/log/auth.log`.
2. **The agent reads it.** The logcollector module of the agent on ubuntu-web reads the new line and sends it to the manager (10.10.10.10:1514, encrypted).
3. **The manager decodes it.** The raw line is split into fields: program (sshd), user, source IP.
4. **A rule matches.** Analysisd checks the rules; 5760 matches and a level 5 alert is created.
5. **The indexer stores it.** Filebeat writes the alert to the `wazuh-alerts-*` index.
6. **The dashboard shows it.** The dashboard reads the alert from the indexer and displays it in Threat Hunting.

An agent first enrols over port **1515**; after that, events flow over **1514**. Not every event becomes an alert: only those that match a rule with a level above 0 reach the dashboard.

6 Security and hygiene S1–S2

The lab is isolated: only siem-host reaches the internet. The Git repository contains no passwords, keys or certificates.

S1 No secrets in Git

Passwords are not written directly in `docker-compose.yml`. Variables such as `${INDEXER_PASSWORD}` are used instead, and the real values live in a `.env` file that never enters Git. Certificates and `internal_users.yml` are also in `.gitignore`:

```
# .gitignore
.env
*.pem
*.key
config/wazuh_indexer_ssl_certs/
config/wazuh_indexer/internal_users.yml
```

Check: `git check-ignore -v .env` confirmed the file is ignored. `git ls-files` lists no `.pem`, `.key` or `.env` files. A search for the new password in `git log -p --all` returned 0 results.

S2 Lab isolation

ubuntu-web and win-client needed the NAT (VMnet8) adapter only during setup, to download the Wazuh agent package, Apache, Sysmon and the SwiftOnSecurity configuration. After installation I removed the NAT adapter from both VMs. They are now only on the VMnet2 lab network.

The proof is in the later screenshots: on ubuntu-web `ip a` shows only `ens33` (10.10.10.50) (Figure 5), and on win-client `ipconfig` shows only `Ethernet0` (10.10.10.80) (Figure 6). Neither has a 192.168.121.x NAT address. Only siem-host keeps a permanent NAT adapter, because it needs it for Docker images and updates.

> **About the gateway:** win-client has 10.10.10.1 set as its default gateway, because Windows would not accept the static IP configuration with the gateway left empty. 10.10.10.1 is the Dell's VMnet2 host-only adapter, not a router. A host-only network does not forward traffic to the internet, so win-client has no route to the internet.

7 Documentation and final status D1–D3

The project is kept in a GitHub repository: README, architecture/, evidence/ and a journal.md with at least five entries.

D1 Repository structure

Path	Contents
`README.md`	Version (Wazuh 4.14.7), start-up order, table of the 5 events, memory table, event path, Windows evaluation expiry date
`architecture/`	Lab diagram and address table (name, IP, RAM, disk)
`evidence/`	The screenshots used in this report
`journal.md`	What I did, what broke, what fixed it
`.gitignore`	Excludes passwords, keys and certificates (S1)

D2 Starting and stopping the lab

Start:

1. Power on siem-host. The containers come up on their own with `restart: always`; check with `docker compose ps` that all three are Up.
2. Wait 1–2 minutes and open the dashboard at `https://10.10.10.10`.
3. Power on ubuntu-web, then win-client; the agents become Active.

Stop:

1. Shut down win-client and ubuntu-web.
2. On siem-host run `docker compose stop` (never `down -v`, which deletes the data volumes).
3. Shut down siem-host.

D3 Journal

#	What I did	What broke	What fixed it
1	Pulled the Wazuh images	"no space left on device" — the LVM volume was only ~18 GB	`growpart`, `pvresize`, `lvextend -r`; root became ~251 GB
2	Changed the admin password	Dashboard returned 401	Wrote the same password in `internal_users.yml` and `docker-compose.yml`
3	Installed the win-client agent	The agent would not connect	Set the manager address to 10.10.10.10
4	Checked the clocks	win-client had the wrong time zone	`Set-TimeZone` to Asia/Baku
5	Reviewed Sysmon events	Most events were "background noise"	Identified those with parent `wazuh-agent.exe` as the agent's own checks
6	Rebooted siem-host	—	Containers, agents and alerts all survived

✓ Final status

Code	Requirement	Status	Evidence
L1–L4	VMs, network, clocks, Docker	Done	Figures 1–9
L5	Snapshots	Done	w1-clean-install, w1-done
W1–W5	Wazuh stack, HTTPS, password, persistence, memory	Done	Figures 10–20
A1–A4	Agents, Sysmon, web server	Done	Figures 21–28
V1	Five known events	Done	Figures 29–34
V2	Event path	Done	Section 5
S1–S2	Git hygiene, lab isolation	Done	Section 6
D1–D3	Repository, README, journal	Done	Section 7